



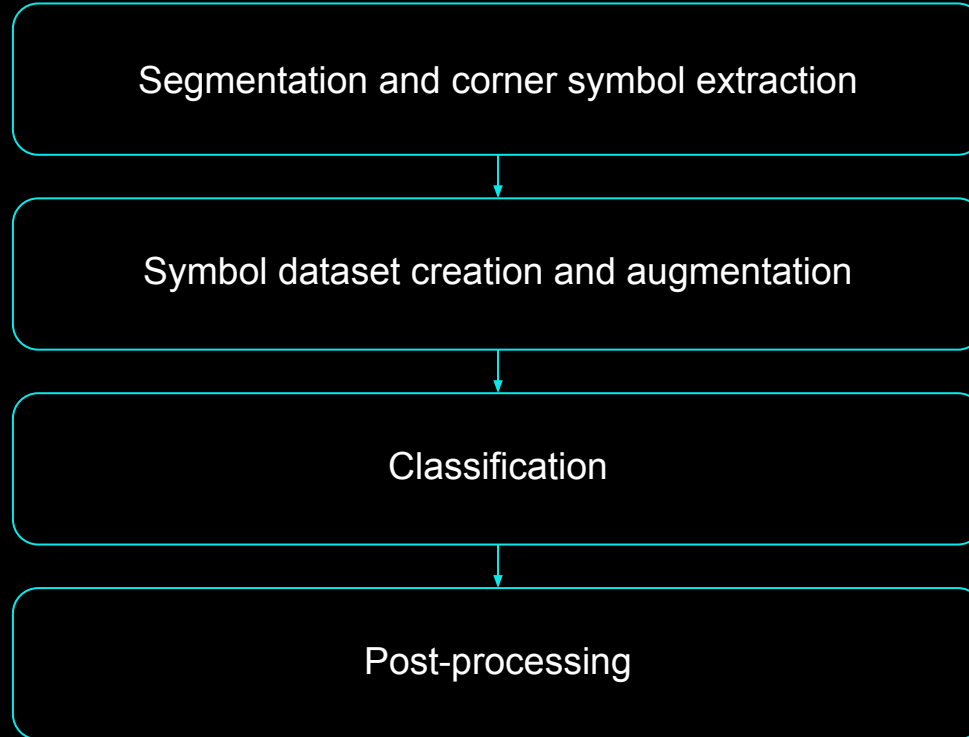
UNO Vision Challenge

Image Analysis and Pattern Recognition



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Grouzinoff Salomé

The Pipeline



Segmentation: the Input

L1000995.jpg



L1000896.jpg



Segmentation: Token Detection

L1000995.jpg: yellow token



L1000896.jpg: grey token



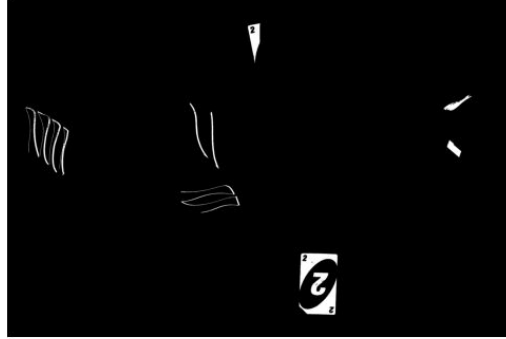
Segmentation: Color Thresholding

Raw color masks after area filtering

Blue



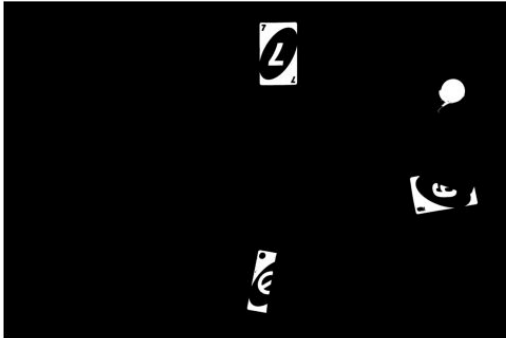
Green



Red



Yellow



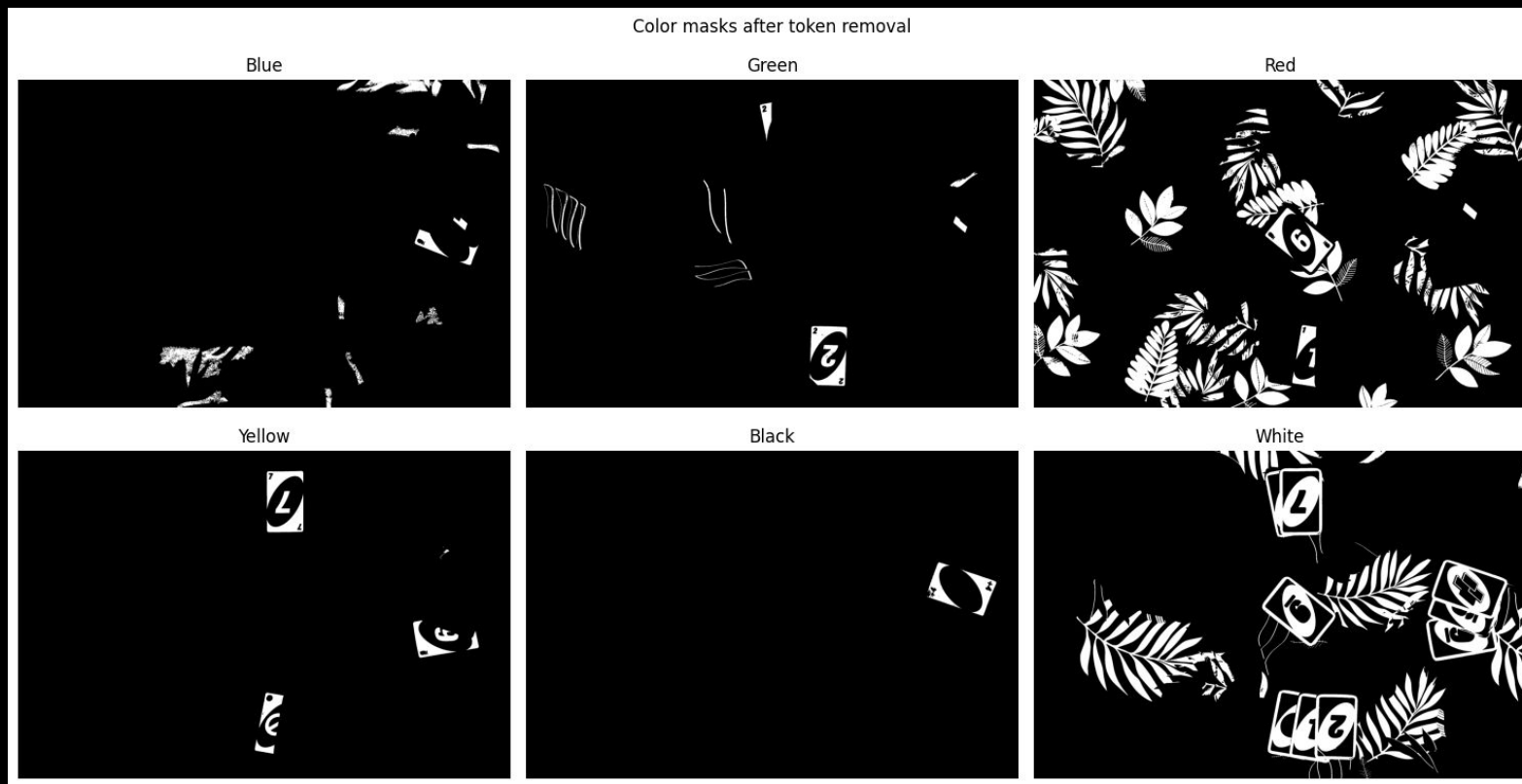
Black



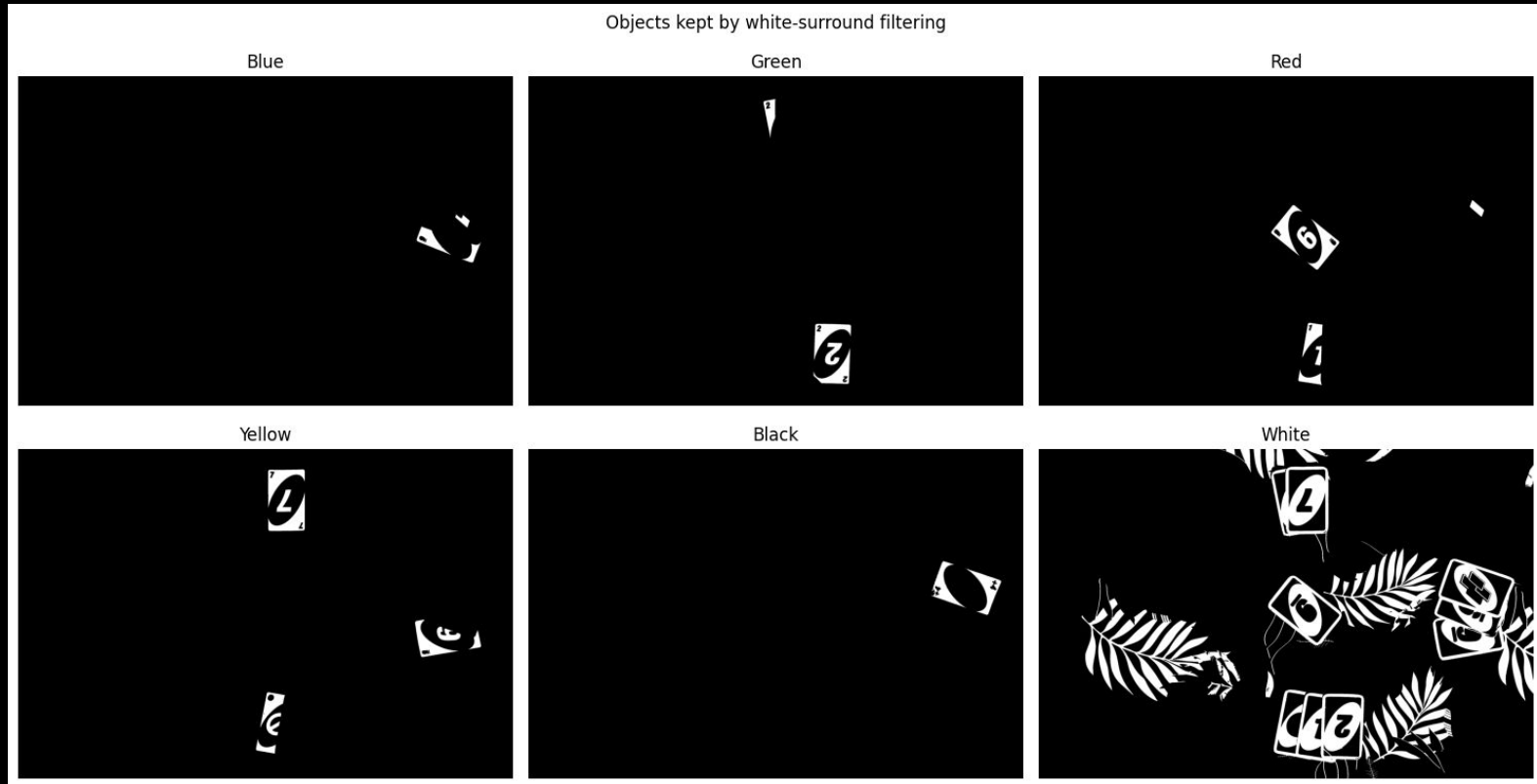
White



Segmentation: Token Removal



Segmentation: Filtering out non-card objects

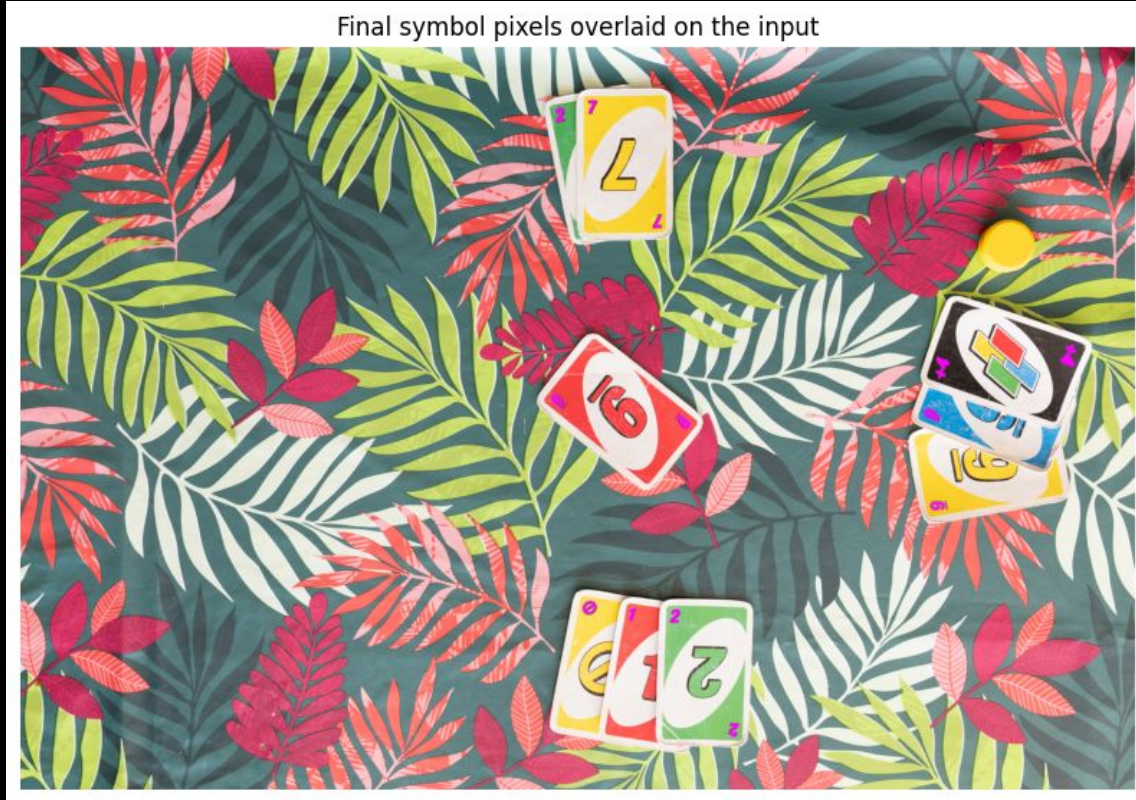


Segmentation: Finding symbols

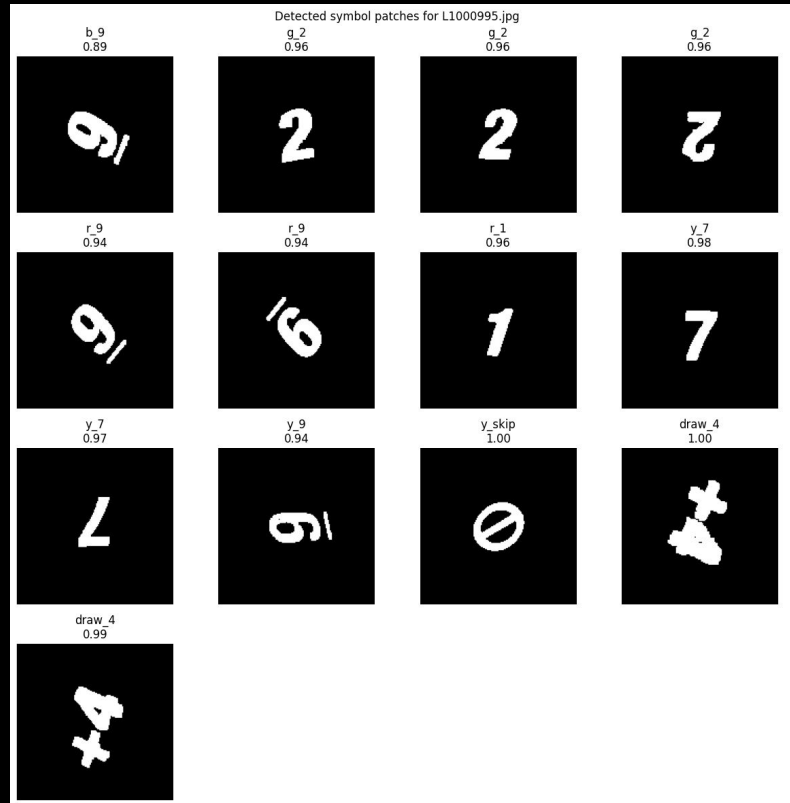
All holes inside colored card regions



Segmentation: Filtering out non-symbol objects



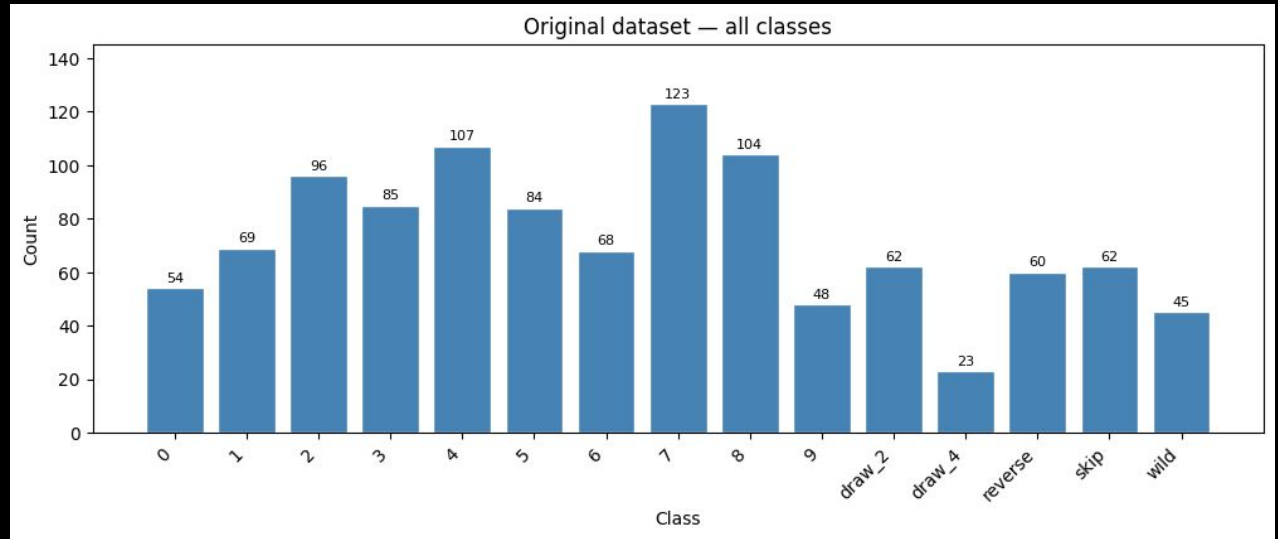
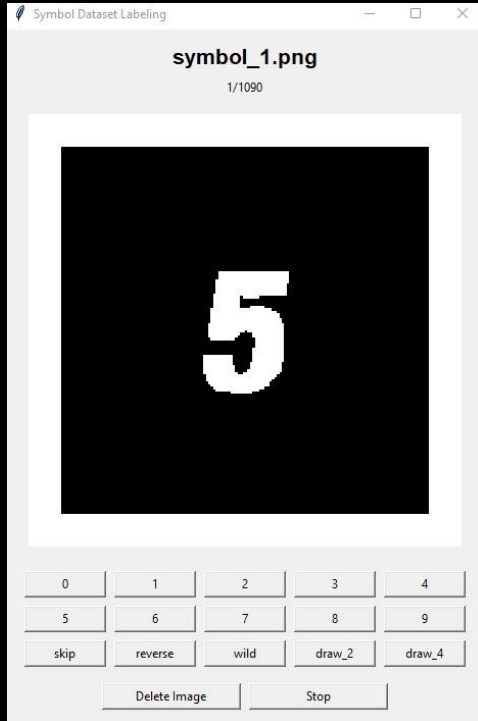
Segmentation: Symbol extraction



Segmentation: Limitations

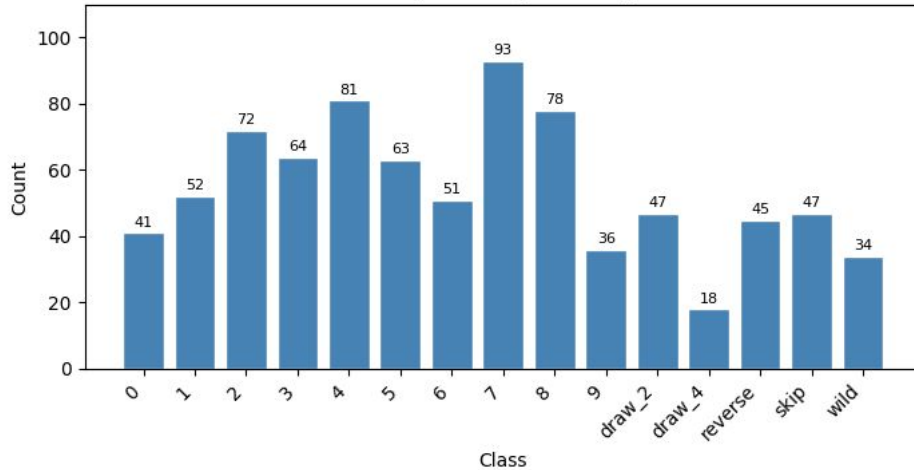


Dataset Preparation for Classification: Building and Labeling

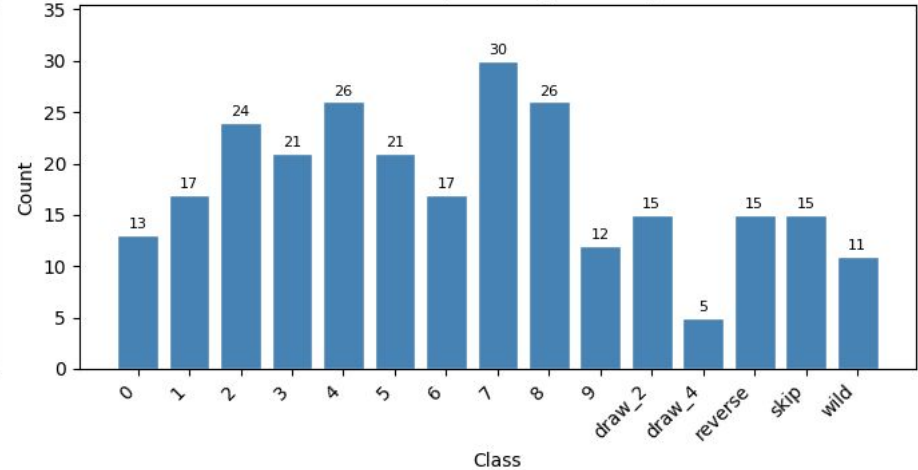


Dataset Preparation for Classification: Train & Validation Split

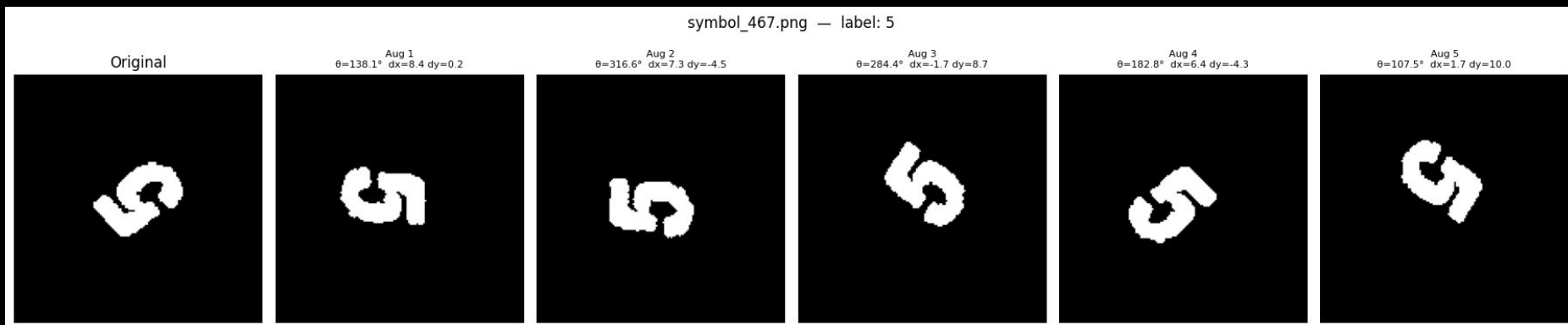
Train split — original images



Validation split — original images

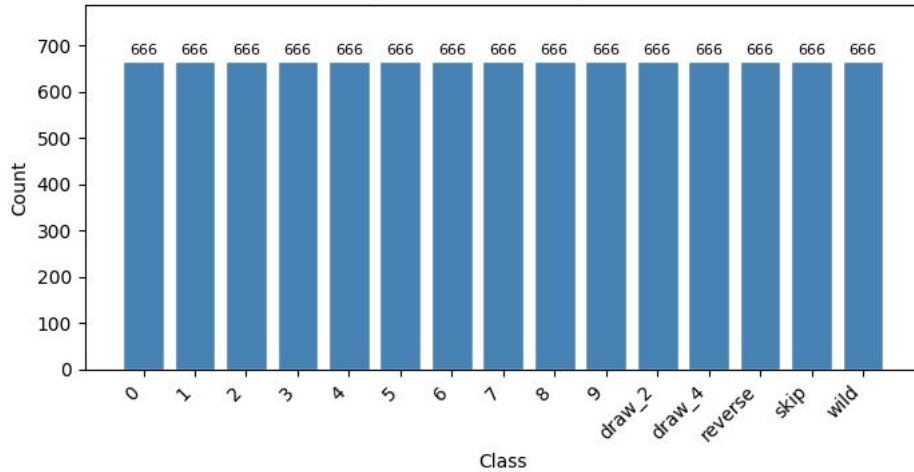


Dataset Preparation for Classification: Augmentation

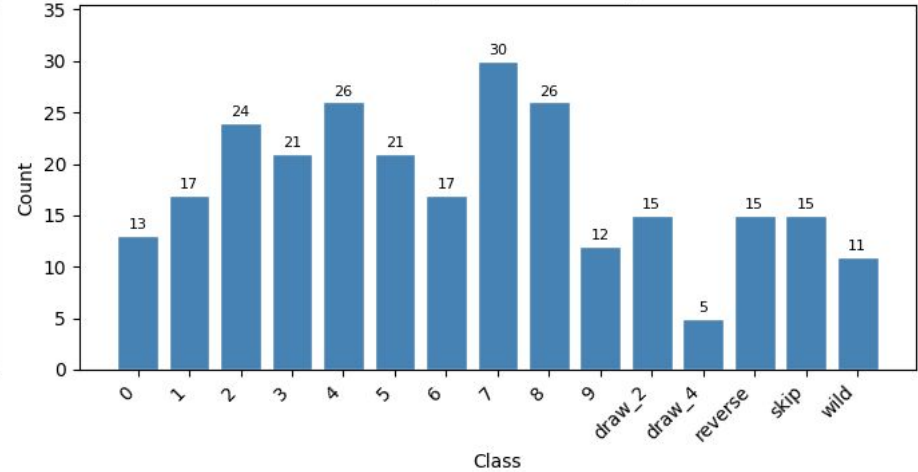


Dataset Preparation for Classification: Final Train & Validation Split

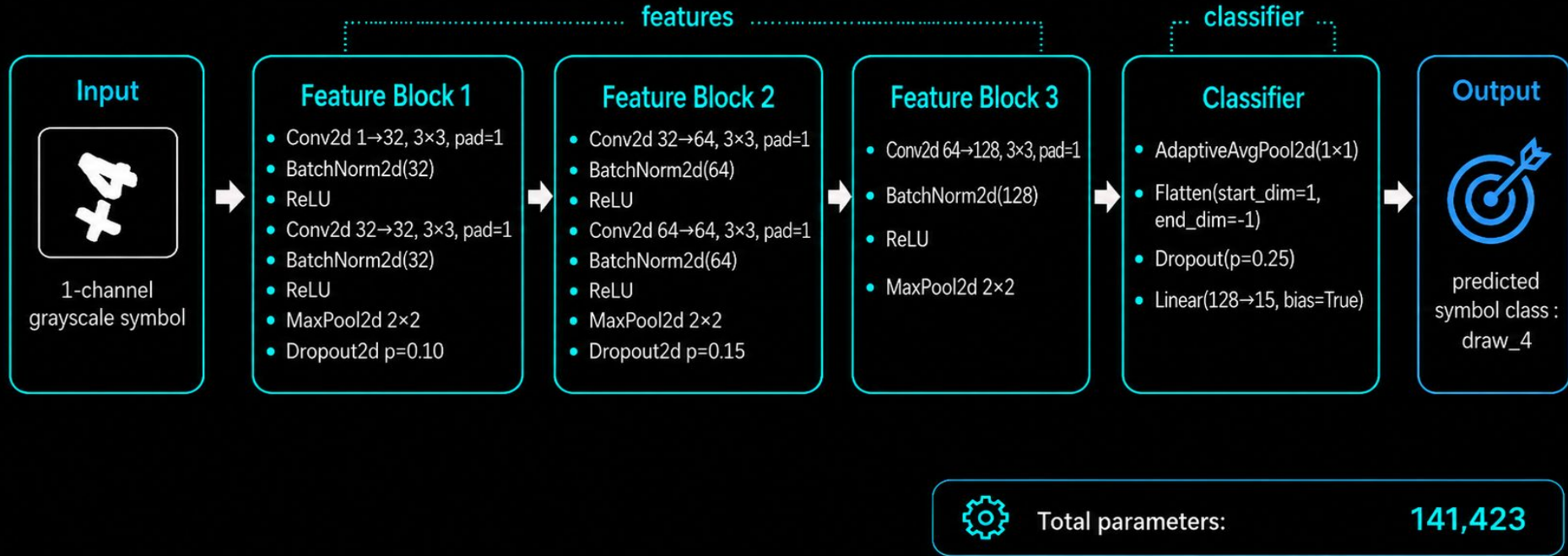
Train split - final after augmentation



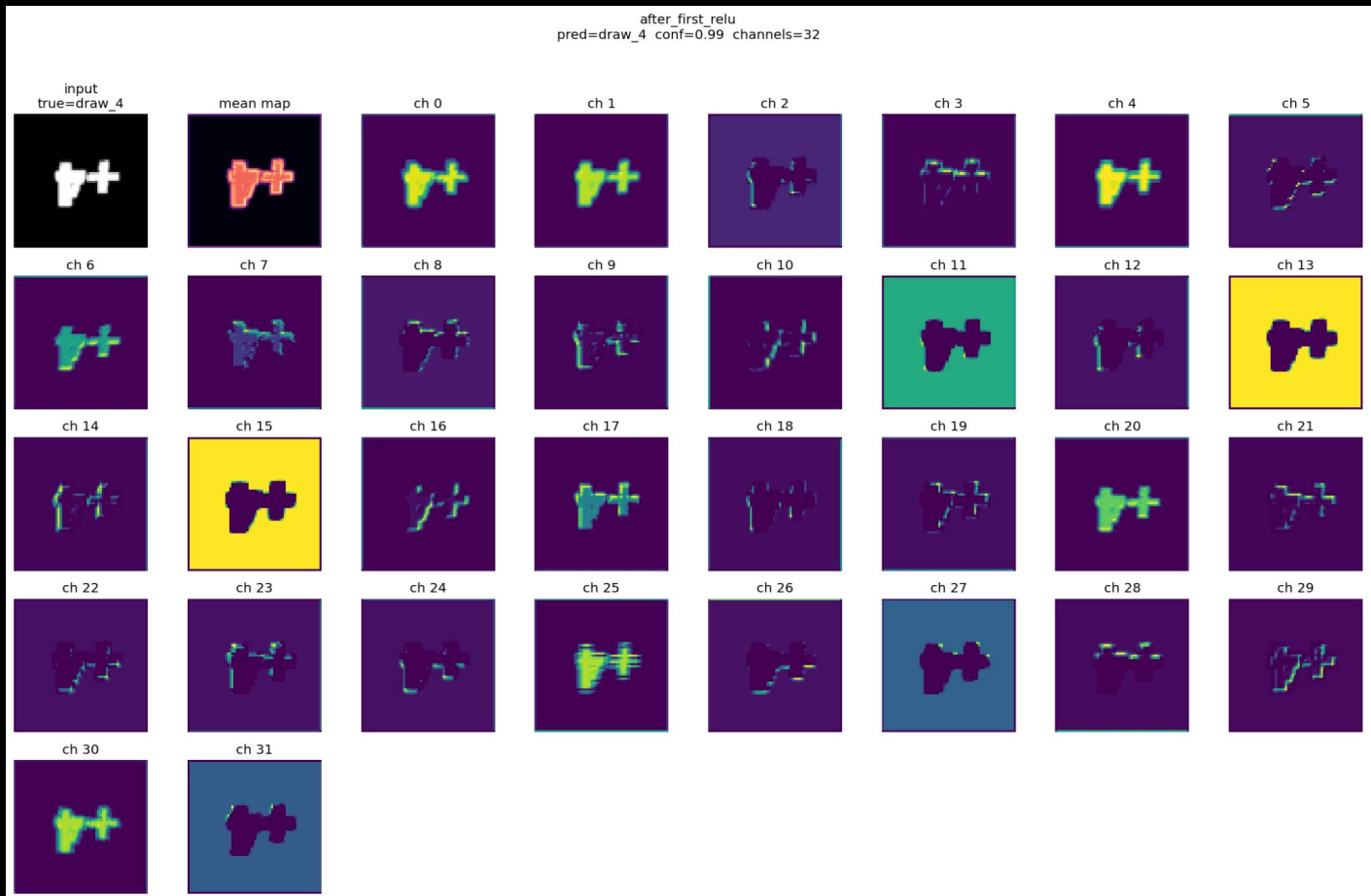
Validation split - held-out originals



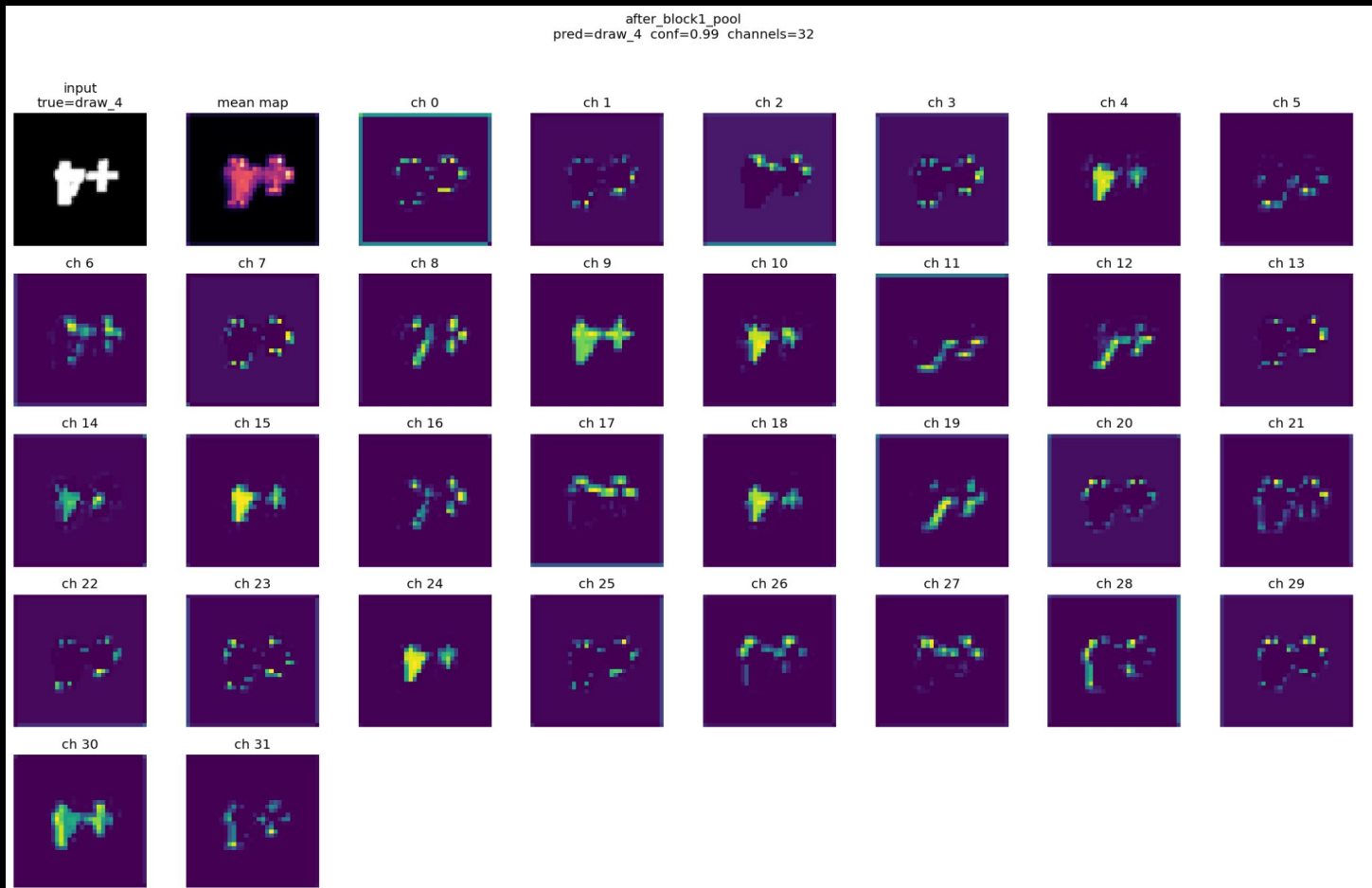
Classification: CNN Architecture



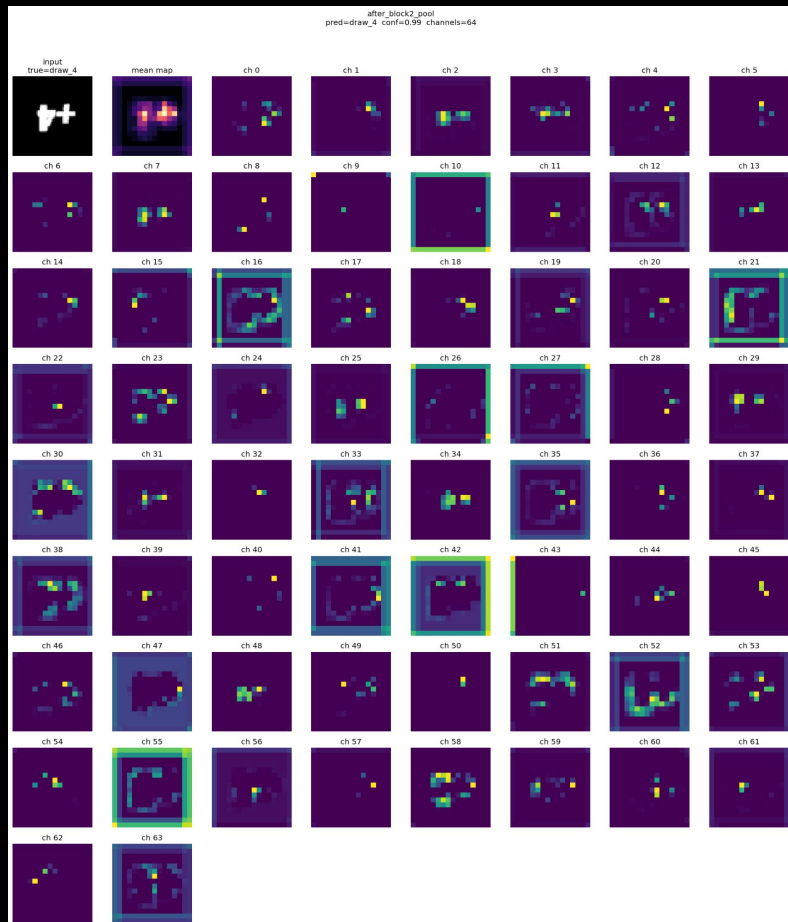
Classification: CNN Architecture



Classification: CNN Architecture

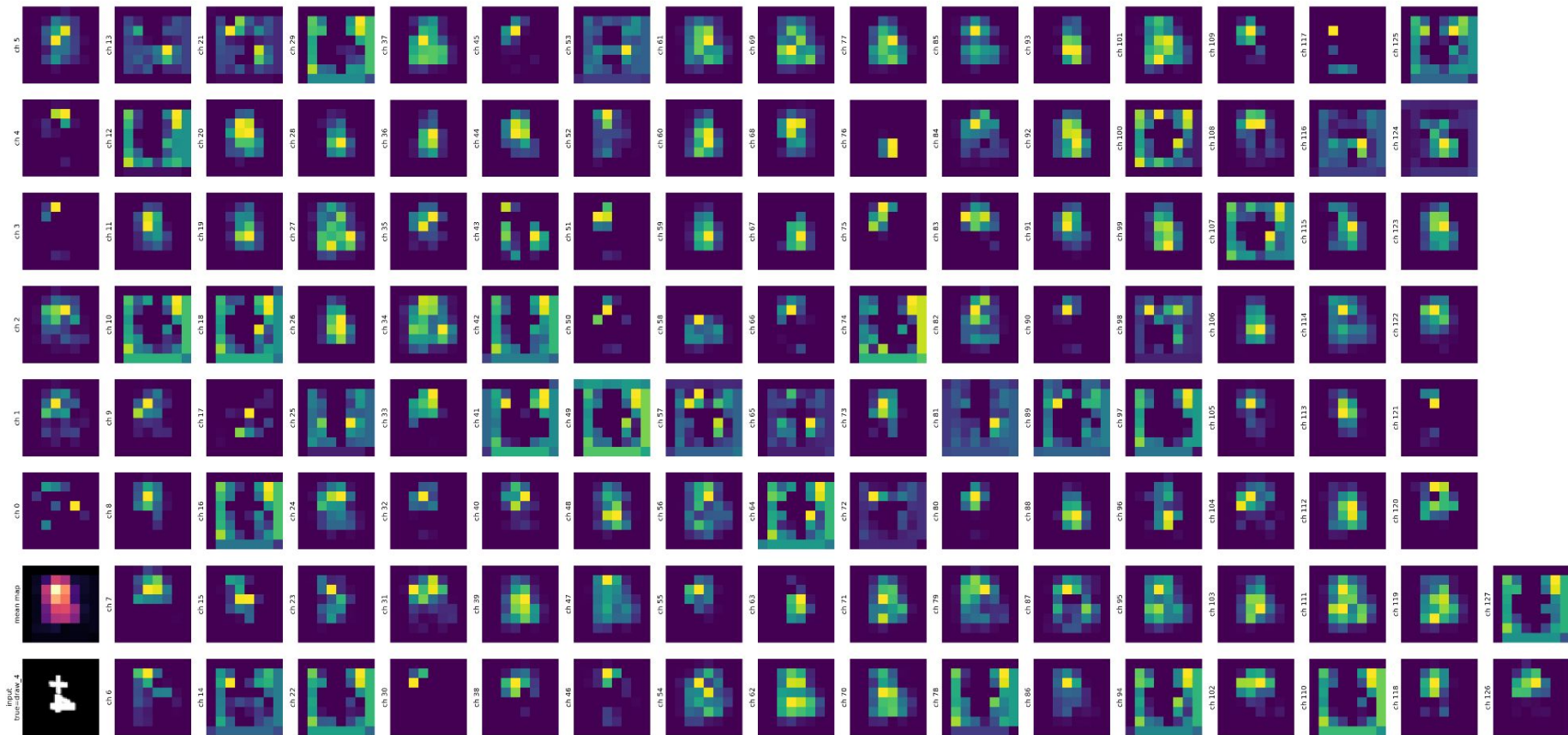


Classification: CNN Architecture

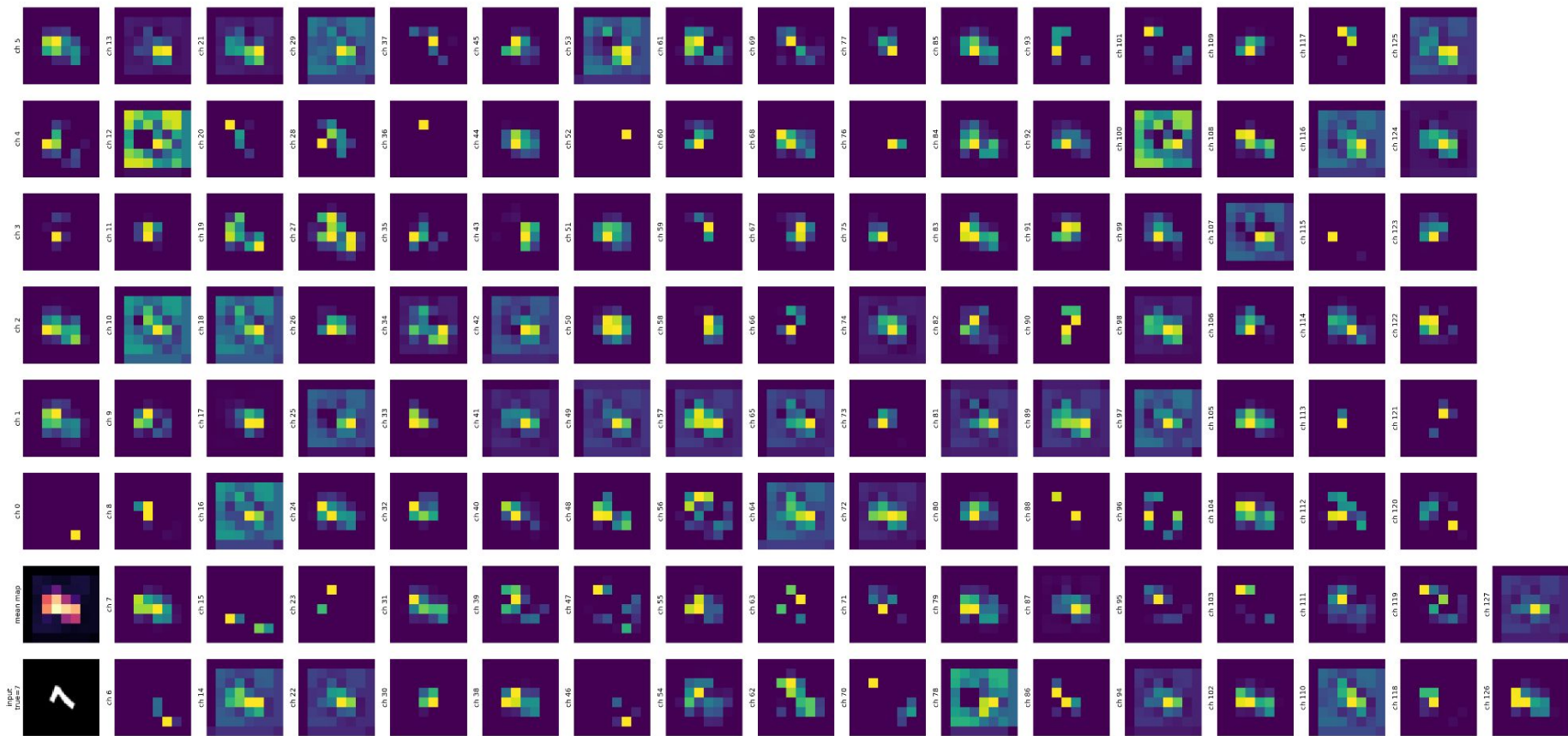


Classification: CNN Architecture

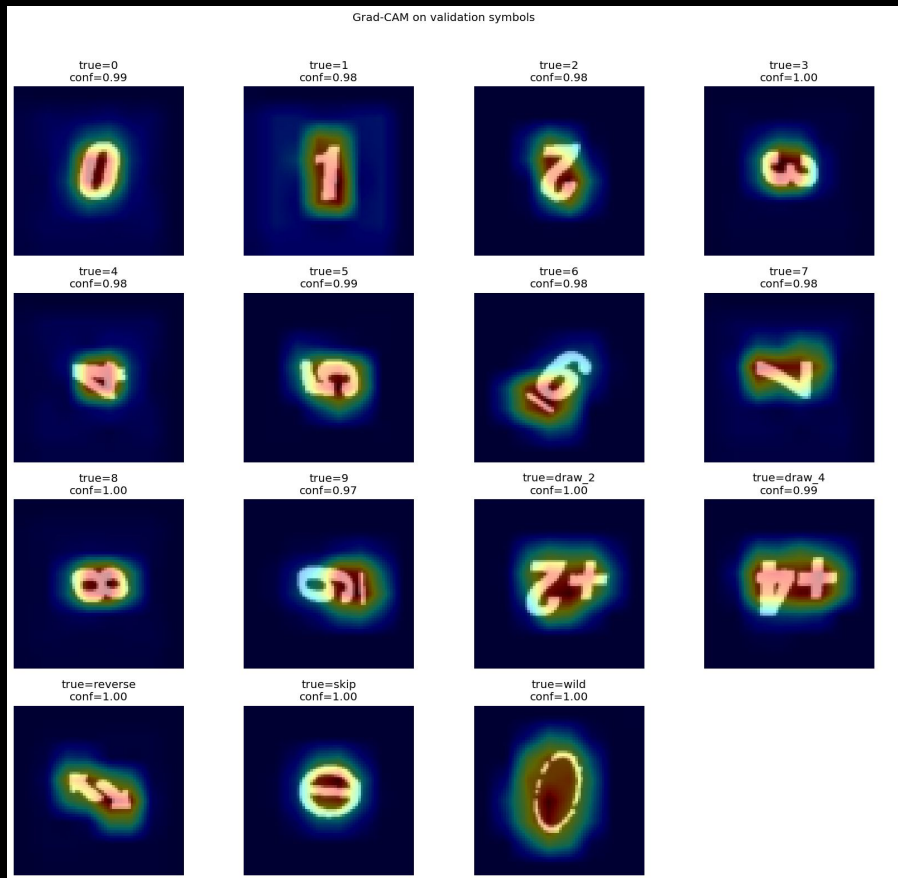
defn: blocks: pool
pred: draw_4 conv: 0.99 Channels: 128



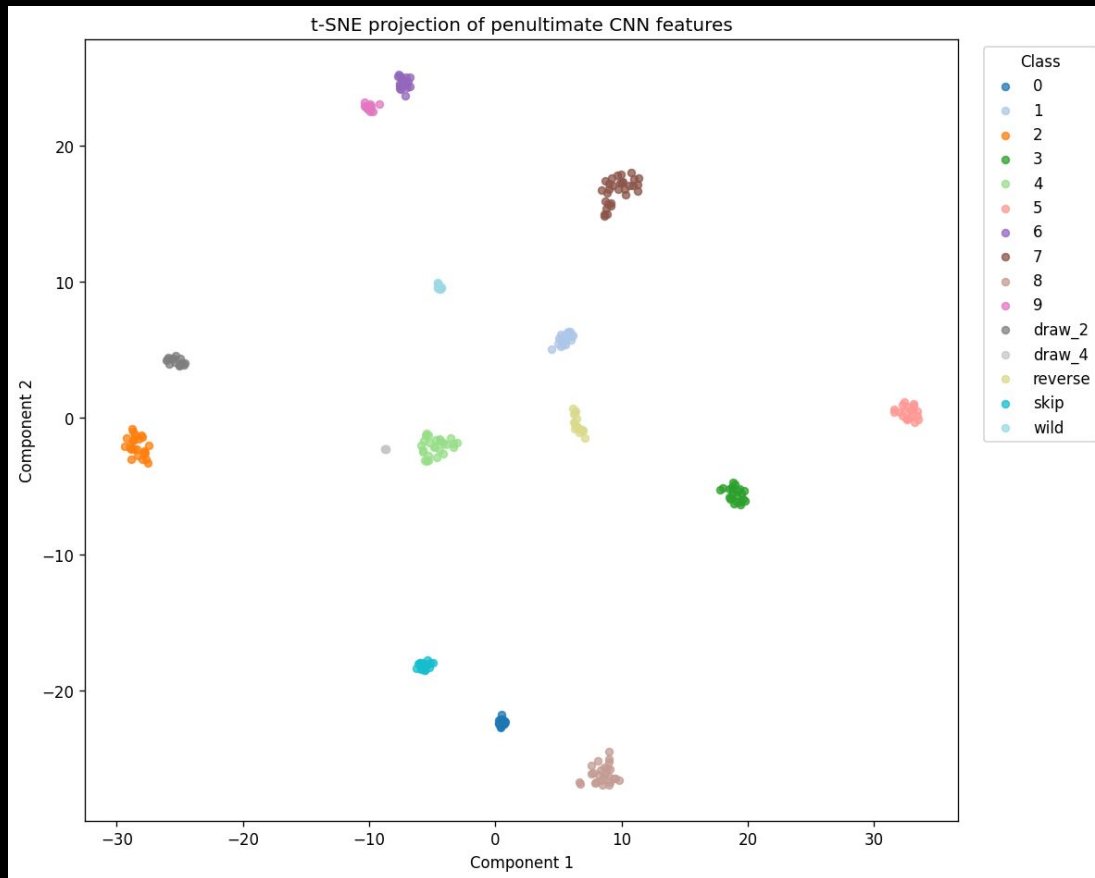
```
after_block3_pool
pred=7 conf=0.91 channels=128
```



Classification: Gradient-weighted Class Activation Mapping

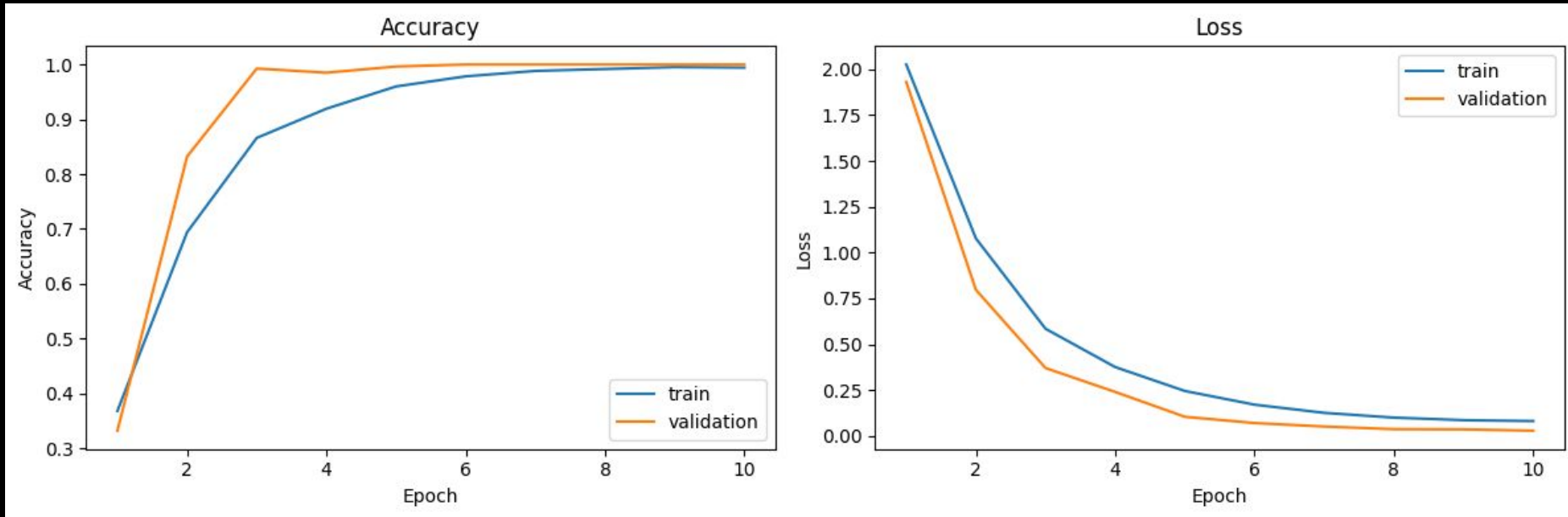


Classification: Feature vector projection



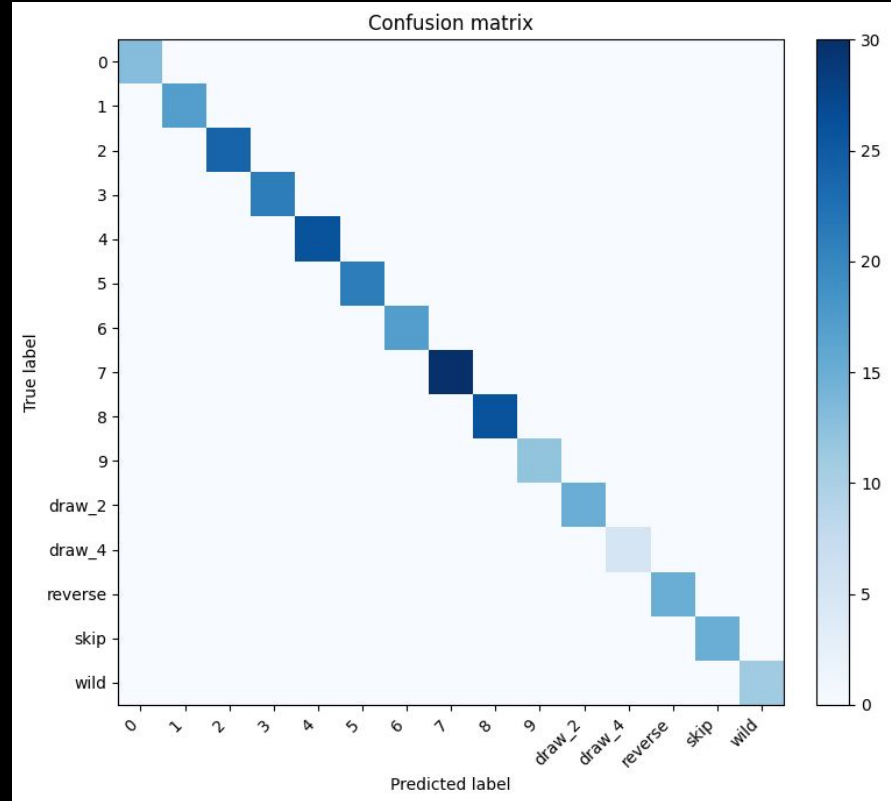
Classification: CNN Training & Validation

Cross-Entropy Loss

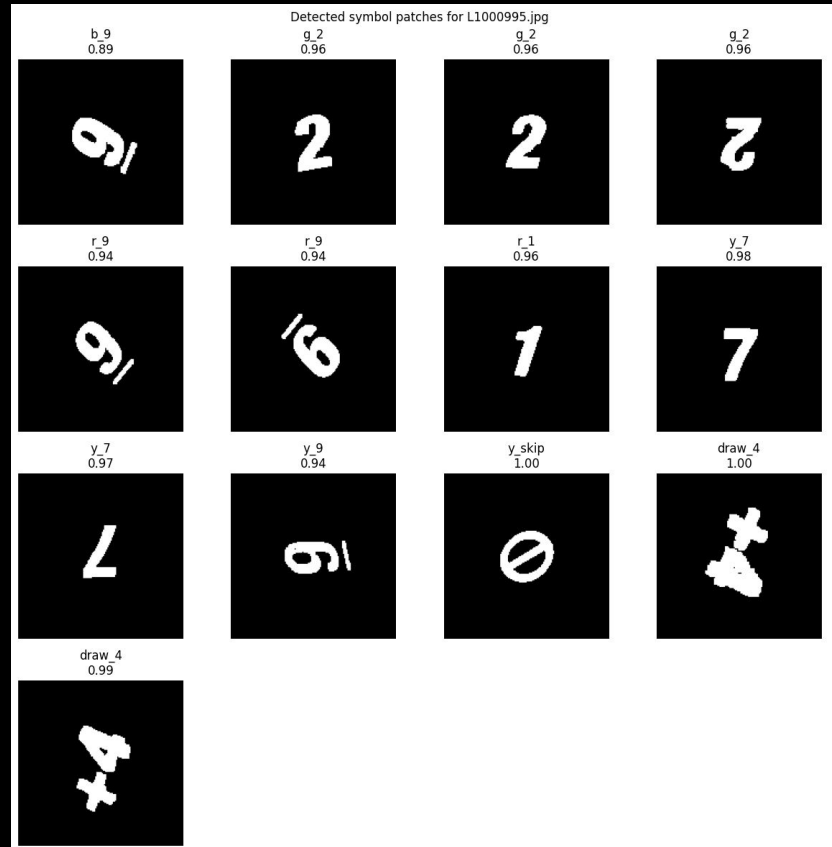


Classification: CNN Training & Validation

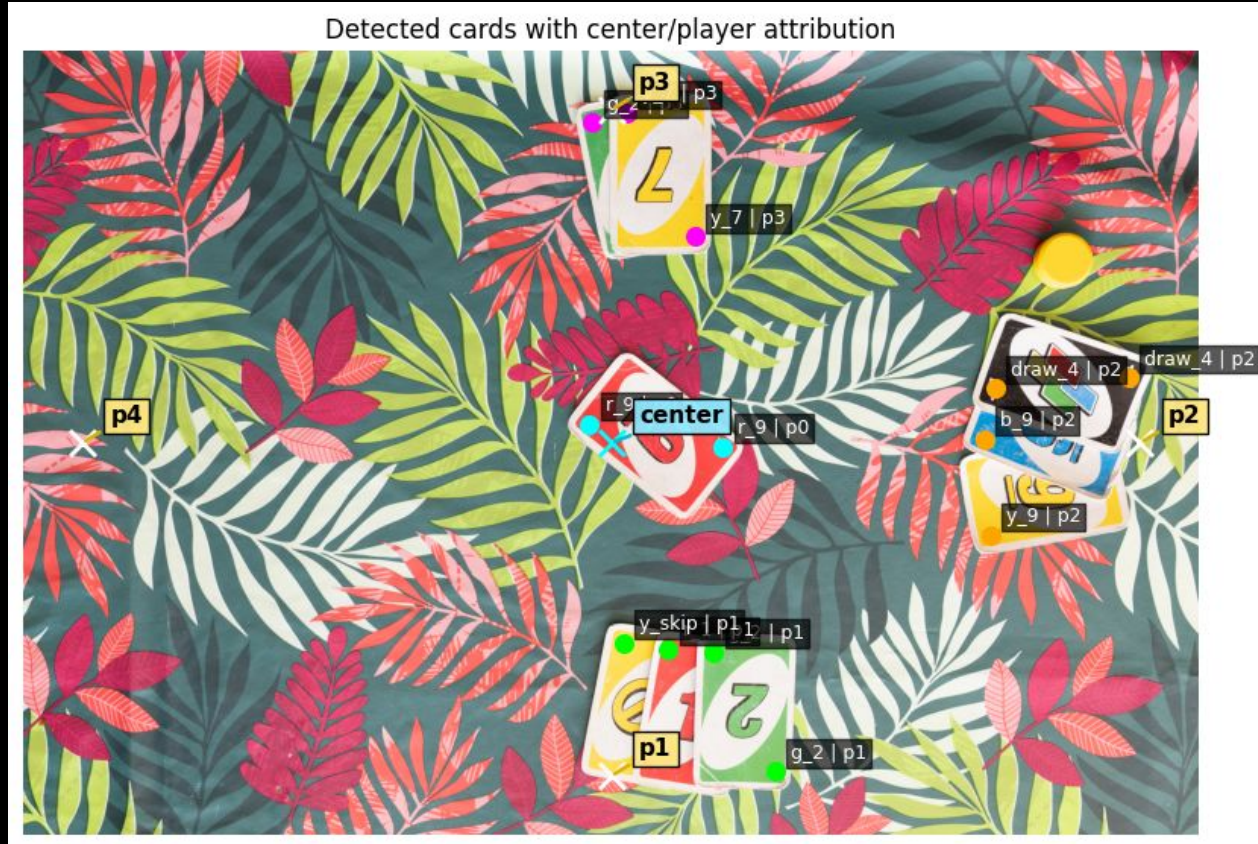
- F1 score : 1.0
- Accuracy : 1.0



Classification: CNN Training & Validation

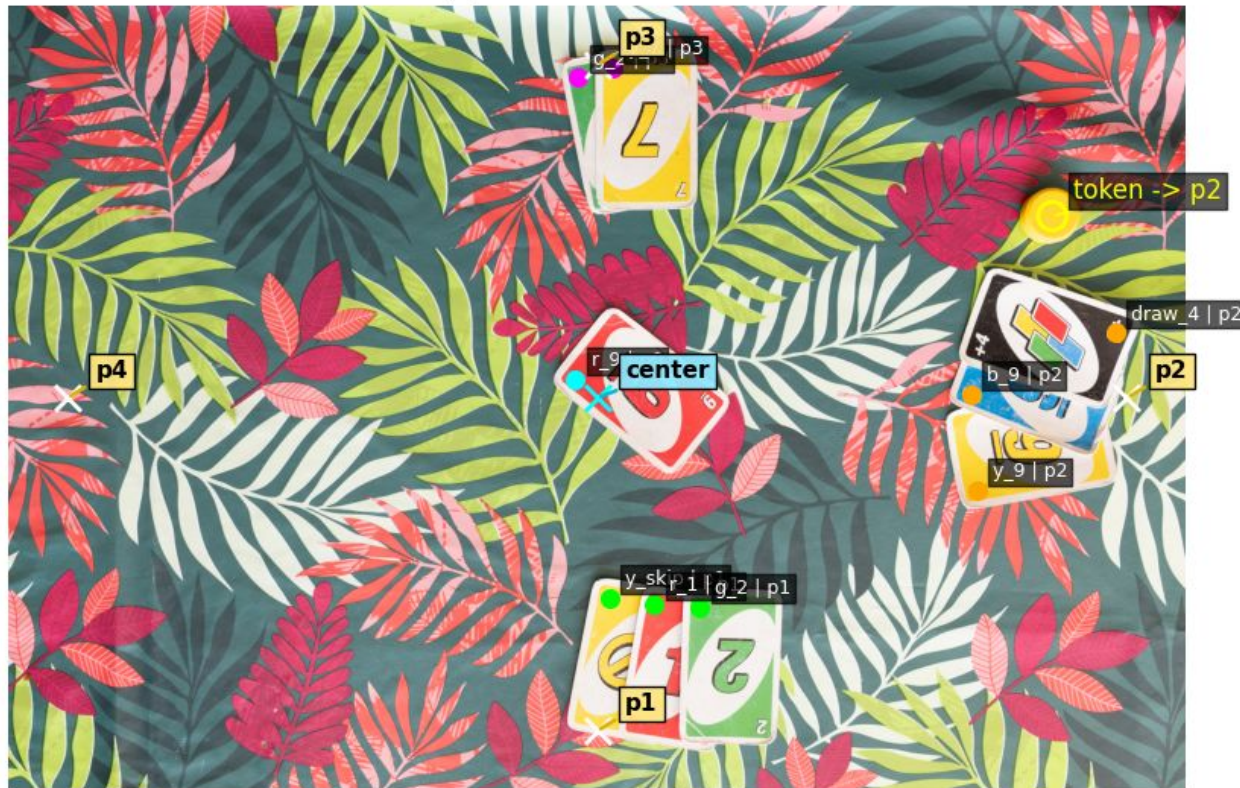


Post-processing: Symbol to player assignment



Post-processing: Final result

Full pipeline result on one frame





**THANK YOU FOR YOUR
ATTENTION**

